

Machine Learning Project Phase II

Wildfire, Smoke and Non-Fire Image Classification Using Convolutional Neural Networks and Transfer Learning

Submitted By

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Course

MSc Data Science

Module

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University

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1. Background and Motivation

Wildfires are among the most destructive natural disasters, causing severe environmental damage, economic loss, and threats to human life. Early detection of fire and smoke is essential for reducing response time and minimizing damage. Traditional monitoring approaches often rely on human observation and sensor-based systems, which may not always provide rapid and accurate detection. Recent advances in Deep Learning and Computer Vision enable automated image-based wildfire detection systems capable of identifying fire and smoke from visual data. This project investigates the use of Convolutional Neural Networks (CNNs) and Transfer Learning for classifying wildfire-related images.

2. Problem Statement

Wildfires represent a major environmental and societal challenge, causing extensive damage to ecosystems, biodiversity, infrastructure, and human life. The increasing frequency and intensity of wildfires due to climate change have highlighted the need for efficient and reliable early detection systems. Traditional wildfire monitoring approaches, such as manual observation, surveillance towers, and sensor-based systems, often suffer from limitations including delayed detection, restricted coverage, and high operational costs.

Recent developments in Artificial Intelligence, Computer Vision, and Deep Learning have created opportunities for automated wildfire detection using image analysis. Convolutional Neural Networks (CNNs) have demonstrated remarkable performance in image classification tasks by automatically learning complex visual features from large-scale image datasets. However, accurately distinguishing between Fire, Smoke, and Non-Fire images remains a challenging task due to variations in lighting conditions, environmental backgrounds, weather effects, and visual similarities between different classes.

This project aims to address this challenge by developing and evaluating a deep learning-based image classification system capable of automatically categorizing images into Fire, Smoke, and Non-Fire classes. Furthermore, the study investigates the effectiveness of a custom CNN architecture and compares its performance with a transfer learning approach based on ResNet50. The findings of this research will contribute to understanding the suitability of deep learning techniques for automated wildfire monitoring and early warning applications.

3. Dataset Information

Dataset Name:

Forest Fire, Smoke and Non-Fire Image Dataset

Total Images:

42,898

Dataset Source:

Kaggle

Dataset Link:

<https://www.kaggle.com/datasets/amerzishminha/forest-fire-smoke-and-non-fire-image-dataset>

Number of Classes:

- Fire
- Smoke
- Non Fire

Image Type:

RGB Images

Training Images:

25,919

Validation Images:

6,479

Testing Images:

10,500

4. Research Questions

1. How effectively can Convolutional Neural Networks classify wildfire-related images into Fire, Smoke, and Non-Fire categories?
2. What is the comparative performance of a custom CNN architecture and a transfer learning model (ResNet50) for wildfire image classification?
3. Which image class (Fire, Smoke, or Non-Fire) presents the greatest classification challenge, and what factors contribute to misclassification?
4. To what extent do image preprocessing and data augmentation techniques improve the robustness of wildfire image classification models?

5. Can deep learning-based image classification systems provide a reliable foundation for automated wildfire monitoring and early detection applications?

5. Research Objectives

- To develop a custom CNN model for wildfire image classification.
- To implement a transfer learning model using ResNet50.
- To evaluate model performance using multiple evaluation metrics.
- To compare the effectiveness of CNN and transfer learning approaches.
- To investigate the applicability of deep learning techniques for wildfire monitoring.

6. Proposed Methodology

Data Preprocessing

- Image resizing (224×224)
- Image normalization
- Data augmentation
 - Rotation
 - Zoom
 - Horizontal Flip

CNN Model

The custom CNN architecture consists of:

- Convolution Layers
- Max Pooling Layers
- Batch Normalization
- Dense Layer
- Dropout Layer
- Softmax Output Layer

Transfer Learning Model

ResNet50 pretrained on ImageNet was used as a comparison model.

7. Evaluation Metrics

The following evaluation metrics were used:

- Accuracy
- Precision
- Recall
- F1 Score

- Confusion Matrix
- Classification Report

8. Experimental Results

Custom CNN Results

Accuracy: 88.30%

Classification Report Summary:

Smoke:

Precision 0.91

Recall 0.84

F1 Score 0.87

Fire:

Precision 0.95

Recall 0.88

F1 Score 0.91

Non Fire:

Precision 0.81

Recall 0.93

F1 Score 0.87

ResNet50 Results

Accuracy: 78.83%

9. Model Comparison

Model	Accuracy
Custom CNN	88.30%
ResNet50	78.83%

The Custom CNN achieved higher classification accuracy than the ResNet50 transfer learning model.

10. Expected Figures and Tables

- Sample Dataset Images
- CNN Architecture
- Accuracy Curve
- Loss Curve
- Confusion Matrix
- Classification Report

- Model Comparison Table

11. Conclusion

- This project successfully developed and evaluated a deep learning-based image classification system for wildfire detection using the Forest Fire, Smoke and Non-Fire Image Dataset. A custom Convolutional Neural Network (CNN) and a transfer learning model based on ResNet50 were implemented and compared to assess their effectiveness in classifying wildfire-related images.
- The experimental results demonstrated that the custom CNN achieved a test accuracy of 88.30%, outperforming the ResNet50 transfer learning model, which achieved an accuracy of 78.83%. The findings indicate that the custom CNN was able to learn dataset-specific visual features more effectively and provide superior classification performance for the given wildfire dataset.
- The study also highlighted the importance of image preprocessing, normalization, and data augmentation techniques in improving model generalization and reducing overfitting. Evaluation metrics including Accuracy, Precision, Recall, F1-Score, Classification Report, and Confusion Matrix analysis confirmed the reliability of the proposed approach.
- Overall, the research demonstrates that deep learning techniques can provide an effective solution for automated wildfire image classification and have strong potential for integration into intelligent environmental monitoring and early warning systems. Future work may explore advanced architectures such as EfficientNet, Vision Transformers (ViT), attention mechanisms, real-time video analysis, and deployment in edge computing environments to further improve detection accuracy and operational efficiency.

12. References

1. TensorFlow Documentation
2. Keras Documentation
3. Kaggle Dataset Repository
4. ResNet50 Research Paper

13. Project Resources

GitHub Repository:

<https://github.com/PRAJWAL2866/Wildfire-Detection-CNN>

Kaggle Notebook:

<https://www.kaggle.com/code/shreyansc/wildfire-notebook/notebook?scriptVersionId=325150269>